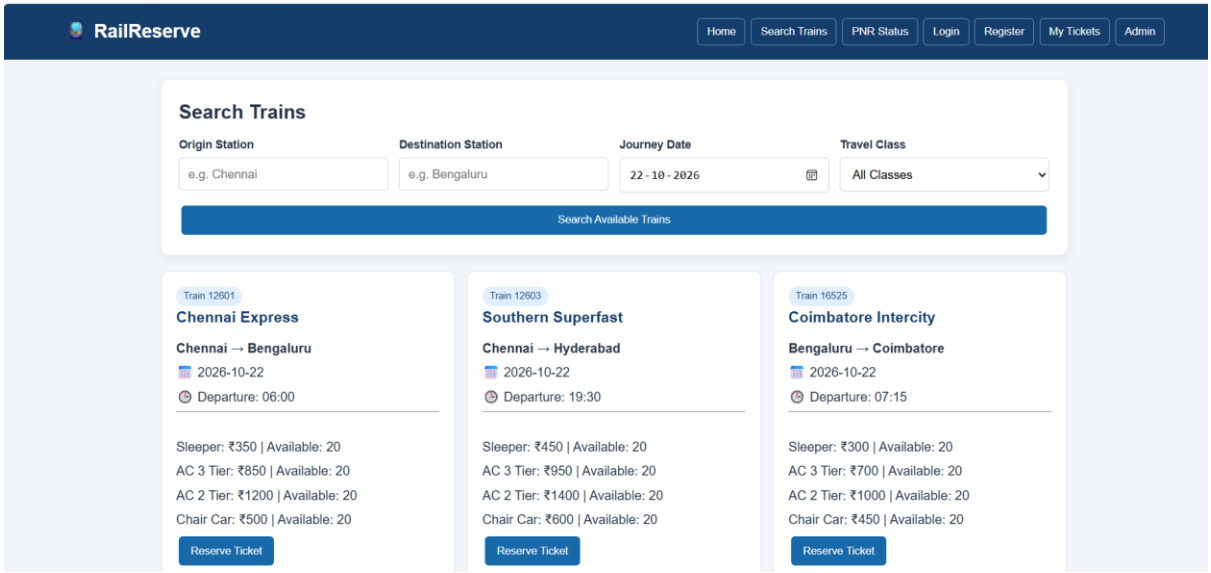
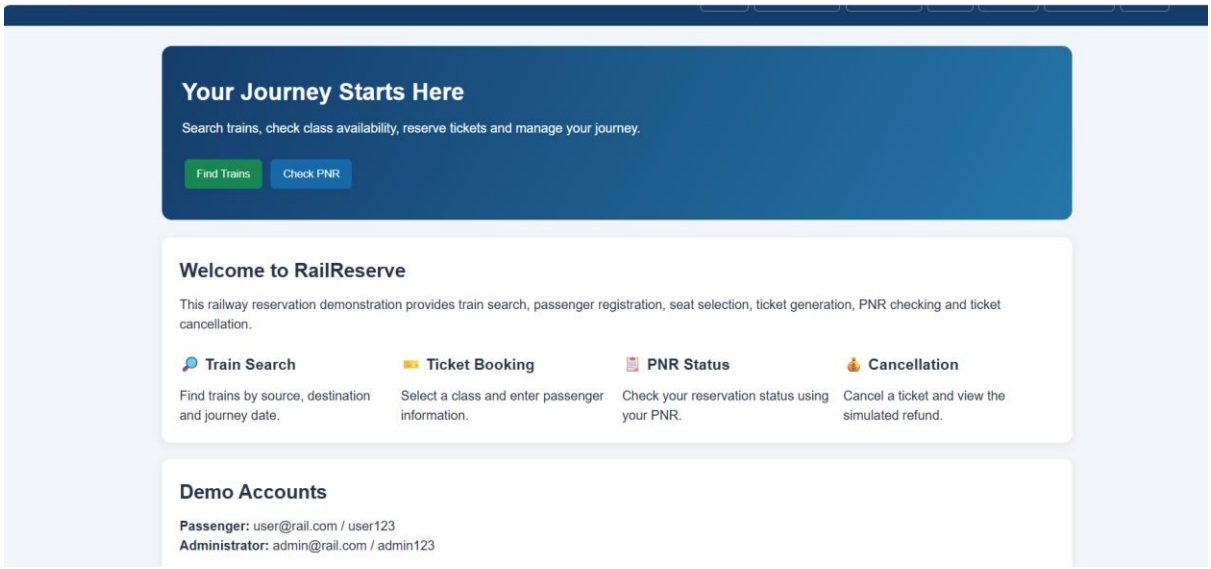
Exp 4- Train reservation system






```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Railway Reservation System</title>


<style>

* {
```

```
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: Arial, sans-serif;
  background: #f2f5f9;
  color: #263449;
  line-height: 1.5;
}

header {
  background: #153d6b;
  color: white;
  padding: 17px 5%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 12px;
}

header h1 {
  font-size: 23px;
}

nav {
```

```
    display: flex;
    flex-wrap: wrap;
    gap: 7px;
}

nav button {
    background: transparent;
    color: white;
    border: 1px solid #ffffff70;
    padding: 8px 11px;
    border-radius: 5px;
    cursor: pointer;
}

nav button:hover {
    background: #285b91;
}

main {
    max-width: 1150px;
    margin: 25px auto;
    padding: 0 15px;
}

.page {
    display: none;
}
```

```
.page.active {
  display: block;
}

.hero {
  background: linear-gradient(120deg, #153d6b, #2576a8);
  color: white;
  padding: 35px 25px;
  border-radius: 12px;
  margin-bottom: 22px;
}

.hero h2 {
  font-size: 29px;
  margin-bottom: 10px;
}

.card {
  background: white;
  padding: 22px;
  border-radius: 10px;
  margin-bottom: 20px;
  box-shadow: 0 3px 12px #152f4b12;
}

h2, h3 {
  margin-bottom: 12px;
}
```

```
form {
  display: grid;
  gap: 13px;
}

.form-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
  gap: 13px;
}

label {
  font-size: 14px;
  font-weight: bold;
}

input, select, textarea {
  width: 100%;
  padding: 11px;
  margin-top: 5px;
  border: 1px solid #cbd5e1;
  border-radius: 5px;
  font-size: 15px;
  background: white;
}

textarea {
```

```
    min-height: 80px;
    resize: vertical;
}

.btn {
    border: none;
    border-radius: 5px;
    padding: 10px 15px;
    margin: 3px;
    background: #1769aa;
    color: white;
    cursor: pointer;
}

.btn:hover {
    background: #104c80;
}

.btn-green {
    background: #198754;
}

.btn-red {
    background: #c0392b;
}

.btn-gray {
    background: #64748b;
```

```
}

.train-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(265px, 1fr));
  gap: 15px;
}

.train-card {
  background: white;
  border: 1px solid #d9e2ed;
  border-radius: 9px;
  padding: 18px;
}

.train-card h3 {
  color: #153d6b;
}

.train-card p {
  margin: 6px 0;
}

.badge {
  display: inline-block;
  padding: 3px 9px;
  border-radius: 20px;
  background: #e1efff;
```

```
    color: #174b80;
    font-size: 12px;
}

.seat-grid {
    display: grid;
    grid-template-columns: repeat(4, minmax(45px, 65px));
    gap: 10px;
    margin: 15px 0;
}

.seat {
    padding: 10px 4px;
    background: #edf2f7;
    border: 1px solid #b9c8d8;
    border-radius: 5px;
    cursor: pointer;
}

.seat.selected {
    background: #198754;
    color: white;
}

.seat.booked {
    background: #cbd5e1;
    color: #64748b;
    cursor: not-allowed;
```

```
}

.status {
  display: inline-block;
  padding: 3px 8px;
  border-radius: 4px;
  background: #d9f7e5;
  color: #155b34;
}

.status.waiting {
  background: #fff0c2;
  color: #7a5200;
}

#message {
  display: none;
  padding: 12px;
  background: #d9f7e5;
  color: #155b34;
  border-radius: 6px;
  margin-bottom: 15px;
}

.table-wrap {
  overflow-x: auto;
}
```

```
table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 12px;
}

th, td {
  padding: 10px;
  border: 1px solid #dbe3ec;
  text-align: left;
}

th {
  background: #e8f0f8;
}

.small {
  font-size: 13px;
  color: #64748b;
}

footer {
  margin-top: 30px;
  padding: 20px;
  background: #153d6b;
  color: white;
  text-align: center;
}
```

```
@media (max-width: 600px) {
  header h1 {
    font-size: 20px;
  }

  nav button {
    padding: 7px;
    font-size: 12px;
  }

  .hero h2 {
    font-size: 23px;
  }

  .card {
    padding: 15px;
  }

  th, td {
    padding: 7px;
    font-size: 12px;
  }
}
</style>
</head>

<body>
```

```
<header>
  <h1>🚆 RailReserve</h1>
  <nav>
    <button onclick="showPage('home')">Home</button>
    <button onclick="showPage('search')">Search Trains</button>
    <button onclick="showPage('pnr')">PNR Status</button>
    <button onclick="showPage('login')">Login</button>
    <button onclick="showPage('register')">Register</button>
    <button onclick="showPage('dashboard')">My Tickets</button>
    <button onclick="showPage('admin')">Admin</button>
  </nav>
</header>

<main>
<div id="message"></div>

<!-- HOME -->
<section id="home" class="page active">
  <div class="hero">
    <h2>Your Journey Starts Here</h2>
    <p>Search trains, check class availability, reserve tickets and manage your journey.</p>
    <br>
    <button class="btn btn-green" onclick="showPage('search')">Find Trains</button>
    <button class="btn" onclick="showPage('pnr')">Check PNR</button>
  </div>
```

```
<div class="card">
  <h2>Welcome to RailReserve</h2>
  <p>This railway reservation demonstration provides train search, passenger registration, seat selection, ticket generation, PNR checking and ticket cancellation.</p>
  <br>
  <div class="form-grid">
    <div>
      <h3>🔎 Train Search</h3>
      <p>Find trains by source, destination and journey date.</p>
    </div>
    <div>
      <h3>🎫 Ticket Booking</h3>
      <p>Select a class and enter passenger information.</p>
    </div>
    <div>
      <h3>📋 PNR Status</h3>
      <p>Check your reservation status using your PNR.</p>
    </div>
    <div>
      <h3>💰 Cancellation</h3>
      <p>Cancel a ticket and view the simulated refund.</p>
    </div>
  </div>
</div>

<div class="card">
  <h2>Demo Accounts</h2>
```

```
      <p><b>Passenger:</b> user@rail.com / user123</p>
      <p><b>Administrator:</b> admin@rail.com / admin123</p>
    </div>
</section>

<!-- TRAIN SEARCH -->
<section id="search" class="page">
  <div class="card">
    <h2>Search Trains</h2>
    <form onsubmit="searchTrains(event)">
      <div class="form-grid">
        <label>Origin Station
          <input id="origin" placeholder="e.g. Chennai" required>
        </label>
        <label>Destination Station
          <input id="destination" placeholder="e.g. Bengaluru" required>
        </label>
        <label>Journey Date
          <input id="journeyDate" type="date" required>
        </label>
        <label>Travel Class
          <select id="classFilter">
            <option value="">All Classes</option>
            <option value="SL">Sleeper (SL)</option>
            <option value="3A">AC 3 Tier (3A)</option>
            <option value="2A">AC 2 Tier (2A)</option>
            <option value="CC">Chair Car (CC)</option>
          </select>
```

```
          </label>
        </div>
        <button type="submit" class="btn">Search Available Trains</button>
      </form>
    </div>

    <div id="trainResults" class="train-grid"></div>
</section>

<!-- BOOKING -->
<section id="booking" class="page">
    <div class="card">
      <h2>Reserve Your Ticket</h2>
      <div id="chosenTrain"></div>

      <h3>Select Travel Class</h3>
      <label>Class
        <select id="travelClass" onchange="updateFareAndSeats()">
          <option value="SL">Sleeper (SL)</option>
          <option value="3A">AC 3 Tier (3A)</option>
          <option value="2A">AC 2 Tier (2A)</option>
          <option value="CC">Chair Car (CC)</option>
        </select>
      </label>

      <p id="fareInfo"></p>
      <h3>Select a Seat</h3>
      <p class="small">Green = selected, grey = already booked.</p>
```

```
<div id="seatLayout" class="seat-grid"></div>
<p id="seatInfo">No seat selected.</p>

<hr>
<br>
<h3>Passenger Details</h3>
<form onsubmit="bookTicket(event)">
  <div class="form-grid">
    <label>Passenger Full Name
      <input id="passengerName" required>
    </label>
    <label>Age
      <input id="passengerAge" type="number" min="1" max="120" required>
    </label>
    <label>Gender
      <select id="passengerGender" required>
        <option value="">Select Gender</option>
        <option>Male</option>
        <option>Female</option>
        <option>Other</option>
      </select>
    </label>
    <label>Contact Email
      <input id="passengerEmail" type="email" required>
    </label>
  </div>

  <label>Payment Mode
```

```
          <select id="paymentMode" required>
            <option value="">Select payment mode</option>
            <option>Online (Demo)</option>
            <option>Offline / Counter (Demo)</option>
          </select>
        </label>

        <p class="small">Demo only: no real money is collected. Selecting a payment mode simulates payment confirmation.</p>
        <button type="submit" class="btn btn-green">Pay & Confirm Ticket</button>
        <button type="button" class="btn btn-gray" onclick="showPage('search')">Back</button>
      </form>
    </div>
  </section>

  <!-- REGISTRATION -->
  <section id="register" class="page">
    <div class="card">
      <h2>Passenger Registration</h2>
      <form onsubmit="registerUser(event)">
        <label>Full Name
          <input id="regName" required>
        </label>
        <label>Email
          <input id="regEmail" type="email" required>
        </label>
        <label>Password
          <input id="regPassword" type="password" minlength="4" required>
```

```
        </label>
        <label>Confirm Password
          <input id="regConfirm" type="password" minlength="4" required>
        </label>
        <button class="btn" type="submit">Create Account</button>
      </form>
      <p>Already registered? <a href="#" onclick="showPage('login')">Log in</a></p>
    </div>
</section>

<!-- LOGIN -->
<section id="login" class="page">
  <div class="card">
    <h2>Passenger / Admin Login</h2>
    <form onsubmit="loginUser(event)">
      <label>Email
        <input id="loginEmail" type="email" required>
      </label>
      <label>Password
        <input id="loginPassword" type="password" required>
      </label>
      <button type="submit" class="btn">Login</button>
      <button type="button" class="btn btn-gray" onclick="forgotPassword()">Forgot Password?</button>
    </form>
    <br>
    <p class="small">Passenger: user@rail.com / user123</p>
    <p class="small">Admin: admin@rail.com / admin123</p>
```

```
  </div>
</section>

<!-- PNR STATUS -->
<section id="pnr" class="page">
  <div class="card">
    <h2>Check PNR Status</h2>
    <form onsubmit="checkPNR(event)">
      <label>Enter PNR Number
        <input id="pnrInput" placeholder="e.g. 700000001" required>
      </label>
      <button class="btn" type="submit">Check Status</button>
    </form>
    <div id="pnrResult"></div>
  </div>
</section>

<!-- PASSENGER DASHBOARD -->
<section id="dashboard" class="page">
  <div class="card">
    <h2>Passenger Dashboard</h2>
    <p id="welcomeUser">Please log in to manage your tickets.</p>
    <button class="btn" onclick="showPage('search')">Book Another Ticket</button>
    <button class="btn btn-gray" onclick="changePassword()">Change
Password</button>
    <button class="btn btn-red" onclick="logout()">Logout</button>
  </div>
```

```
  <div class="card">
    <h3>My Reservation History</h3>
    <div id="myTickets" class="table-wrap"></div>
  </div>
</section>

<!-- TICKET -->
<section id="ticket" class="page">
  <div class="card" id="ticketContent"></div>
  <button class="btn" onclick="window.print()">Print Ticket</button>
  <button class="btn btn-gray" onclick="showPage('dashboard')">Back to Dashboard</button>
</section>

<!-- ADMIN -->
<section id="admin" class="page">
  <div class="card">
    <h2>Railway Administration</h2>
    <p id="adminStatus">Please log in with administrator credentials.</p>
    <button class="btn" onclick="showPage('login')">Admin Login</button>
  </div>

  <div id="adminTools" style="display:none">
    <div class="card">
      <h3>Add / Update Train Details</h3>
      <form onsubmit="saveTrain(event)">
        <input id="editTrainId" type="hidden">
        <div class="form-grid">
```

```
<label>Train Number
  <input id="trainNumber" required>
</label>
<label>Train Name
  <input id="trainName" required>
</label>
<label>Origin Station
  <input id="trainOrigin" required>
</label>
<label>Destination Station
  <input id="trainDestination" required>
</label>
<label>Departure Time
  <input id="trainTime" type="time" required>
</label>
<label>Journey Date
  <input id="trainDate" type="date" required>
</label>
<label>Sleeper Fare (₹)
  <input id="fareSL" type="number" min="1" required>
</label>
<label>AC 3 Tier Fare (₹)
  <input id="fare3A" type="number" min="1" required>
</label>
<label>AC 2 Tier Fare (₹)
  <input id="fare2A" type="number" min="1" required>
</label>
<label>Chair Car Fare (₹)
```

```
            <input id="fareCC" type="number" min="1" required>
          </label>
          <label>Seats per Class
            <input id="trainSeats" type="number" min="1" max="40" required>
          </label>
        </div>
        <button type="submit" class="btn btn-green">Save Train</button>
        <button type="button" class="btn btn-gray"
onclick="clearTrainForm()">Clear</button>
      </form>
    </div>

    <div class="card">
      <h3>Manage Train Schedules</h3>
      <div id="allTrains" class="table-wrap"></div>
    </div>

    <div class="card">
      <h3>Manage Registered Users</h3>
      <div id="allUsers" class="table-wrap"></div>
    </div>

    <div class="card">
      <h3>Reservation and Payment Reports</h3>
      <button class="btn" onclick="generateReport()">Generate Report</button>
      <button class="btn btn-gray" onclick="downloadReport()">Download
CSV</button>
      <div id="reportOutput"></div>
    </div>
```

```
      <div class="card">
        <h3>Refund Rules</h3>
        <p>Set the refund percentage for cancelled tickets.</p>
        <label>Refund Percentage (%)
          <input id="refundPercent" type="number" min="0" max="100" value="80">
        </label>
        <button class="btn" onclick="saveRefundRule()">Update Refund Rule</button>
        <p class="small">This is a configurable classroom example, not an official railway refund policy.</p>
      </div>
    </div>
</section>

<!-- CONTACT -->
<section id="contact" class="page">
  <div class="card">
    <h2>Contact Us</h2>
    <form onsubmit="contactUs(event)">
      <label>Name <input id="contactName" required></label>
      <label>Email <input id="contactEmail" type="email" required></label>
      <label>Message <textarea id="contactMessage" required></textarea></label>
      <button type="submit" class="btn">Send Message</button>
    </form>
  </div>
</section>
</main>
```

```
<footer>
  <p>© 2026 RailReserve | Internet Programming Lab Experiment</p>
  <p>Train Search · Reservations · PNR Status · Administration</p>
  <button class="btn" onclick="showPage('contact')">Contact Us</button>
</footer>

<script>
let users = [
  {name: "Demo Passenger", email: "user@rail.com", password: "user123", role: "user"},
  {name: "Railway Admin", email: "admin@rail.com", password: "admin123", role: "admin"}
];

let trains = [
  {
    id: 1, number: "12601", name: "Chennai Express",
    origin: "Chennai", destination: "Bengaluru", time: "06:00",
    date: "", seats: 20,
    fares: {SL: 350, "3A": 850, "2A": 1200, CC: 500}
  },
  {
    id: 2, number: "12603", name: "Southern Superfast",
    origin: "Chennai", destination: "Hyderabad", time: "19:30",
    date: "", seats: 20,
    fares: {SL: 450, "3A": 950, "2A": 1400, CC: 600}
  },
  {
    id: 3, number: "16525", name: "Coimbatore Intercity",
```

```
    origin: "Bengaluru", destination: "Coimbatore", time: "07:15",
    date: "", seats: 20,
    fares: {SL: 300, "3A": 700, "2A": 1000, CC: 450}
  }
];

let reservations = [];
let currentUser = null;
let chosenTrainId = null;
let chosenSeat = null;
let nextTrainId = 4;
let nextPNR = 700000001;
let refundPercentage = 80;

const classNames = {
  SL: "Sleeper",
  "3A": "AC 3 Tier",
  "2A": "AC 2 Tier",
  CC: "Chair Car"
};

function localDate() {
  const d = new Date();
  return d.getFullYear() + "-" +
    String(d.getMonth() + 1).padStart(2, "0") + "-" +
    String(d.getDate()).padStart(2, "0");
}
```

```
const today = localDate();
trains.forEach(t => t.date = today);

document.getElementById("journeyDate").value = today;
document.getElementById("journeyDate").min = today;
document.getElementById("trainDate").value = today;
document.getElementById("trainDate").min = today;

function safe(value) {
  return String(value ?? "").replace(/[&<>"']/g, ch => ({
    "&": "&amp;", "<": "&lt;", ">": "&gt;",
    '"': "&quot;", "'": "&#39;"
  }[ch]));
}

function showMessage(text) {
  const box = document.getElementById("message");
  box.textContent = text;
  box.style.display = "block";
  window.scrollTo({top: 0, behavior: "smooth"});
}

function showPage(id) {
  document.querySelectorAll(".page").forEach(p => p.classList.remove("active"));
  document.getElementById(id).classList.add("active");
  document.getElementById("message").style.display = "none";

  if (id === "dashboard") renderDashboard();
```

```
  if (id === "admin") renderAdmin();
  if (id === "search") renderTrains(trains);

  window.scrollTo(0, 0);
}

function searchTrains(event) {
  event.preventDefault();

  const from = document.getElementById("origin").value.trim().toLowerCase();
  const to = document.getElementById("destination").value.trim().toLowerCase();
  const date = document.getElementById("journeyDate").value;
  const cls = document.getElementById("classFilter").value;

  const results = trains.filter(t =>
    t.origin.toLowerCase() === from &&
    t.destination.toLowerCase() === to &&
    t.date === date &&
    (!cls || t.fares[cls] !== undefined)
  );

  renderTrains(results);

  if (!results.length) {
    document.getElementById("trainResults").innerHTML =
      "<div class='card'>No trains found for the selected route and date. Try another search.</div>";
  }
```

```
}

function availableSeats(train, cls) {
  const used = reservations.filter(r =>
    r.trainId === train.id &&
    r.travelClass === cls &&
    r.status !== "Cancelled"
  ).length;

  return Math.max(0, train.seats - used);
}

function renderTrains(list) {
  const area = document.getElementById("trainResults");

  if (!list.length) {
    area.innerHTML = "<div class='card'>No train schedules are currently listed.</div>";
    return;
  }

  area.innerHTML = list.map(t => `
    <div class="train-card">
      <span class="badge">Train ${safe(t.number)}</span>
      <h3>${safe(t.name)}</h3>
      <p><b>${safe(t.origin)} → ${safe(t.destination)}</b></p>
      <p>🗓️ ${t.date}</p>
      <p>🕒 Departure: ${t.time}</p>
      <hr><br>
```

```
        <p>Sleeper: ₹${t.fares.SL} | Available: ${availableSeats(t, "SL")}</p>
        <p>AC 3 Tier: ₹${t.fares["3A"]} | Available: ${availableSeats(t, "3A")}</p>
        <p>AC 2 Tier: ₹${t.fares["2A"]} | Available: ${availableSeats(t, "2A")}</p>
        <p>Chair Car: ₹${t.fares.CC} | Available: ${availableSeats(t, "CC")}</p>
        <button class="btn" onclick="startReservation(${t.id})">Reserve Ticket</button>
      </div>
    `).join("");
}

function startReservation(id) {
  if (!currentUser || currentUser.role !== "user") {
    alert("Please log in with a passenger account before reserving a ticket.");
    showPage("login");
    return;
  }

  chosenTrainId = id;
  chosenSeat = null;

  const t = trains.find(x => x.id === id);
  if (!t) return;

  document.getElementById("chosenTrain").innerHTML = `
    <h3>${safe(t.name)} (${safe(t.number)})</h3>
    <p>${safe(t.origin)} → ${safe(t.destination)}</p>
    <p>Journey date: ${t.date} | Departure: ${t.time}</p>`;

  document.getElementById("passengerName").value = currentUser.name;
```

```
  document.getElementById("passengerEmail").value = currentUser.email;

  updateFareAndSeats();
  showPage("booking");
}

function updateFareAndSeats() {
  const t = trains.find(x => x.id === chosenTrainId);
  if (!t) return;

  const cls = document.getElementById("travelClass").value;

  document.getElementById("fareInfo").textContent =
    "Fare: ₹" + t.fares[cls] +
    " | Available seats: " + availableSeats(t, cls);

  chosenSeat = null;
  renderSeats();
}

function renderSeats() {
  const t = trains.find(x => x.id === chosenTrainId);
  if (!t) return;

  const cls = document.getElementById("travelClass").value;

  const booked = reservations
    .filter(r => r.trainId === t.id &&
```

```
            r.travelClass === cls &&
            r.status !== "Cancelled")
        .map(r => r.seat);

    let html = "";

    for (let i = 1; i <= t.seats; i++) {
        const seat = cls + "-" + i;
        const unavailable = booked.includes(seat);

        html += `<button type="button"
            class="seat ${unavailable ? "booked" : ""} ${chosenSeat === seat ? "selected" : ""}"
            ${unavailable ? "disabled" : ""}
            onclick="chooseSeat('${seat}')">${seat}</button>`;
    }

    document.getElementById("seatLayout").innerHTML = html;
    document.getElementById("seatInfo").textContent =
        chosenSeat ? "Selected seat: " + chosenSeat : "No seat selected.";
}

function chooseSeat(seat) {
    chosenSeat = seat;
    renderSeats();
}

function bookTicket(event) {
    event.preventDefault();
```

```
if (!currentUser || currentUser.role !== "user") {
   alert("Passenger login is required.");
   showPage("login");
   return;
}

if (!chosenSeat) {
   alert("Please select an available seat.");
   return;
}

const t = trains.find(x => x.id === chosenTrainId);
const cls = document.getElementById("travelClass").value;

if (!t || availableSeats(t, cls) <= 0) {
   alert("No seats are available in this class.");
   return;
}

const seatTaken = reservations.some(r =>
   r.trainId === t.id &&
   r.travelClass === cls &&
   r.seat === chosenSeat &&
   r.status !== "Cancelled"
);

if (seatTaken) {
```

```
    alert("That seat is no longer available. Please select another.");
    chosenSeat = null;
    renderSeats();
    return;
}

const payment = document.getElementById("paymentMode").value;
if (!payment) {
    alert("Select a payment mode.");
    return;
}

const reservation = {
    pnr: String(nextPNR++),
    userEmail: currentUser.email,
    trainId: t.id,
    trainNumber: t.number,
    trainName: t.name,
    origin: t.origin,
    destination: t.destination,
    date: t.date,
    time: t.time,
    travelClass: cls,
    seat: chosenSeat,
    passengerName: document.getElementById("passengerName").value.trim(),
    age: document.getElementById("passengerAge").value,
    gender: document.getElementById("passengerGender").value,
    email: document.getElementById("passengerEmail").value.trim(),
```

```
    fare: t.fares[cls],
    paymentMode: payment,
    paymentStatus: "Simulated successful",
    status: "Confirmed",
    refund: 0
  };

  reservations.push(reservation);

  alert("Reservation successful!\nPNR: " + reservation.pnr +
    "\nSeat: " + reservation.seat +
    "\nFare: ₹" + reservation.fare +
    "\nPayment: " + payment +
    "\n\nDemo confirmation only. No real payment or email was processed.");

  chosenSeat = null;
  showPage("ticket");
  renderTicket(reservation);
}

function renderTicket(r) {
  document.getElementById("ticketContent").innerHTML = `
    <h2>🎫 Railway Reservation Ticket</h2>
    <h3>PNR Number: ${safe(r.pnr)}</h3>
    <p><span class="status">${safe(r.status)}</span></p>
    <hr><br>
    <p><b>Passenger:</b> ${safe(r.passengerName)}</p>
    <p><b>Age / Gender:</b> ${safe(r.age)} / ${safe(r.gender)}</p>
```

```
    <p><b>Contact Email:</b> ${safe(r.email)}</p>
    <br>
    <p><b>Train:</b> ${safe(r.trainName)} (${safe(r.trainNumber)})</p>
    <p><b>Route:</b> ${safe(r.origin)} → ${safe(r.destination)}</p>
    <p><b>Journey Date:</b> ${safe(r.date)}</p>
    <p><b>Departure:</b> ${safe(r.time)}</p>
    <p><b>Class:</b> ${safe(classNames[r.travelClass])} (${safe(r.travelClass)})</p>
    <p><b>Seat:</b> ${safe(r.seat)}</p>
    <p><b>Fare:</b> ₹${r.fare}</p>
    <p><b>Payment Mode:</b> ${safe(r.paymentMode)}</p>
    <p><b>Payment Status:</b> ${safe(r.paymentStatus)}</p>
    <hr><br>
    <p class="small">This is a demonstration ticket and is not valid for real railway travel.</p>`;
}

function checkPNR(event) {
  event.preventDefault();

  const pnr = document.getElementById("pnrInput").value.trim();
  const r = reservations.find(x => x.pnr === pnr);
  const area = document.getElementById("pnrResult");

  if (!r) {
    area.innerHTML = "<p>No reservation found for this PNR in the current session.</p>";
    return;
  }
```

```
  area.innerHTML = `
    <div class="card">
      <h3>PNR: ${safe(r.pnr)}</h3>
      <p><b>Passenger:</b> ${safe(r.passengerName)}</p>
      <p><b>Train:</b> ${safe(r.trainName)}</p>
      <p><b>Route:</b> ${safe(r.origin)} → ${safe(r.destination)}</p>
      <p><b>Date:</b> ${safe(r.date)}</p>
      <p><b>Class:</b> ${safe(classNames[r.travelClass])}</p>
      <p><b>Seat:</b> ${safe(r.seat)}</p>
      <p><b>Status:</b> <span class="status ${r.status === "Cancelled" ? "waiting" : ""}">${safe(r.status)}</span></p>
      ${r.status === "Cancelled" ? `<p>Refund amount: ₹${r.refund}</p>` : ""}
      <button class="btn" onclick="openTicket('${safe(r.pnr)}')">View Ticket</button>
    </div>`;
}

function openTicket(pnr) {
  const r = reservations.find(x => x.pnr === pnr);

  if (!r) return;

  if (currentUser && currentUser.role === "user" && r.userEmail !== currentUser.email) {
    alert("You can only view your own ticket.");
    return;
  }

  showPage("ticket");
  renderTicket(r);
```

```
}

function renderDashboard() {
  if (!currentUser || currentUser.role !== "user") {
    document.getElementById("welcomeUser").textContent =
      "Please log in with a passenger account to view your reservations.";
    document.getElementById("myTickets").innerHTML =
      "<p>Your tickets will appear here after you log in.</p>";
    return;
  }

  document.getElementById("welcomeUser").textContent =
    "Welcome, " + currentUser.name + " (" + currentUser.email + ")";

  const mine = reservations.filter(r => r.userEmail === currentUser.email);

  if (!mine.length) {
    document.getElementById("myTickets").innerHTML =
      "<p>No reservations yet. Search for a train to get started.</p>";
    return;
  }

  document.getElementById("myTickets").innerHTML = `
    <table>
      <tr>
        <th>PNR</th><th>Train</th><th>Date</th>
        <th>Class / Seat</th><th>Fare</th><th>Status</th><th>Actions</th>
      </tr>
```

```
        ${mine.map(r => `<tr>
          <td>${safe(r.pnr)}</td>
          <td>${safe(r.trainName)}<br>${safe(r.origin)} → ${safe(r.destination)}</td>
          <td>${safe(r.date)}</td>
          <td>${safe(r.travelClass)} / ${safe(r.seat)}</td>
          <td>₹${r.fare}</td>
          <td>${safe(r.status)}</td>
          <td>
            <button class="btn" onclick="openTicket('${safe(r.pnr)}')">View / Print</button>
            ${r.status !== "Cancelled"
              ? `<button class="btn btn-red" onclick="cancelTicket('${safe(r.pnr)}')">Cancel</button>`
              : `Refund: ₹${r.refund}`}
          </td>
        </tr>`).join("")}
      </table>`;
}

function cancelTicket(pnr) {
  if (!currentUser || currentUser.role !== "user") {
    alert("Passenger login is required.");
    return;
  }

  const r = reservations.find(x => x.pnr === pnr);

  if (!r || r.userEmail !== currentUser.email) {
    alert("You can only cancel your own reservations.");
```

```
    return;
  }

  if (r.status === "Cancelled") {
    alert("This ticket is already cancelled.");
    return;
  }

  if (!confirm("Cancel ticket with PNR " + pnr + "?")) return;

  r.status = "Cancelled";
  r.refund = Math.round(r.fare * refundPercentage / 100);

  alert("Ticket cancelled.\nPNR: " + r.pnr +
      "\nCalculated refund: ₹" + r.refund +
      "\nThis demo calculates the refund but does not transfer money.");

  renderDashboard();
}

function registerUser(event) {
  event.preventDefault();

  const name = document.getElementById("regName").value.trim();
  const email = document.getElementById("regEmail").value.trim().toLowerCase();
  const password = document.getElementById("regPassword").value;
  const confirmPassword = document.getElementById("regConfirm").value;
```

```
  if (password !== confirmPassword) {
    alert("Passwords do not match.");
    return;
  }

  if (users.some(u => u.email.toLowerCase() === email)) {
    alert("This email is already registered.");
    return;
  }

  users.push({name, email, password, role: "user"});
  event.target.reset();
  alert("Registration successful. Please log in.");
  showPage("login");
}

function loginUser(event) {
  event.preventDefault();

  const email = document.getElementById("loginEmail").value.trim().toLowerCase();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(u =>
    u.email.toLowerCase() === email && u.password === password
  );

  if (!user) {
    alert("Invalid email or password.");
```

```
    return;
  }

  currentUser = user;
  alert("Welcome, " + user.name + "!");

  showPage(user.role === "admin" ? "admin" : "dashboard");
}

function logout() {
  currentUser = null;
  showPage("home");
  showMessage("You have logged out.");
}

function forgotPassword() {
  const email = prompt("Enter your registered email:");

  if (!email) return;

  const user = users.find(u =>
    u.email.toLowerCase() === email.trim().toLowerCase()
  );

  if (!user) {
    alert("No account found with that email.");
    return;
  }
```

```
  const password = prompt("Enter a new password (at least 4 characters):");

  if (!password || password.length < 4) {
    alert("Password must contain at least 4 characters.");
    return;
  }

  user.password = password;
  alert("Password updated for this demo session. No email was sent.");
}

function changePassword() {
  if (!currentUser) {
    alert("Please log in first.");
    return;
  }

  const oldPassword = prompt("Enter your current password:");

  if (oldPassword !== currentUser.password) {
    alert("Incorrect current password.");
    return;
  }

  const newPassword = prompt("Enter new password (at least 4 characters):");

  if (!newPassword || newPassword.length < 4) {
```

```
    alert("Password must contain at least 4 characters.");
    return;
  }

  currentUser.password = newPassword;
  alert("Password changed for this session.");
}

function isAdmin() {
  return currentUser && currentUser.role === "admin";
}

function renderAdmin() {
  if (!isAdmin()) {
    document.getElementById("adminTools").style.display = "none";
    document.getElementById("adminStatus").textContent =
      "Log in with admin@rail.com / admin123 to access the administration panel.";
    return;
  }

  document.getElementById("adminTools").style.display = "block";
  document.getElementById("adminStatus").textContent =
    "Administrator logged in: " + currentUser.email;

  document.getElementById("allTrains").innerHTML = `
    <table>
      <tr><th>Number</th><th>Name</th><th>Route</th><th>Date / Time</th><th>Actions</th></tr>
```

```
        ${trains.map(t => `<tr>
          <td>${safe(t.number)}</td>
          <td>${safe(t.name)}</td>
          <td>${safe(t.origin)} → ${safe(t.destination)}</td>
          <td>${safe(t.date)} / ${safe(t.time)}</td>
          <td>
            <button class="btn" onclick="editTrain(${t.id})">Edit</button>
            <button class="btn btn-red" onclick="deleteTrain(${t.id})">Delete</button>
          </td>
        </tr>`).join("")}
      </table>`;

    document.getElementById("allUsers").innerHTML = `
      <table>
        <tr><th>Name</th><th>Email</th><th>Role</th><th>Action</th></tr>
        ${users.map((u, i) => `<tr>
          <td>${safe(u.name)}</td>
          <td>${safe(u.email)}</td>
          <td>${safe(u.role)}</td>
          <td>${u.role === "user"
            ? `<button class="btn btn-red" onclick="removeUser(${i})">Remove</button>`
            : "Protected"}</td>
        </tr>`).join("")}
      </table>`;
}

function saveTrain(event) {
```

```
event.preventDefault();

if (!isAdmin()) {
  alert("Admin login required.");
  return;
}

const id = Number(document.getElementById("editTrainId").value);

const data = {
  number: document.getElementById("trainNumber").value.trim(),
  name: document.getElementById("trainName").value.trim(),
  origin: document.getElementById("trainOrigin").value.trim(),
  destination: document.getElementById("trainDestination").value.trim(),
  time: document.getElementById("trainTime").value,
  date: document.getElementById("trainDate").value,
  seats: Number(document.getElementById("trainSeats").value),
  fares: {
    SL: Number(document.getElementById("fareSL").value),
    "3A": Number(document.getElementById("fare3A").value),
    "2A": Number(document.getElementById("fare2A").value),
    CC: Number(document.getElementById("fareCC").value)
  }
};

if (data.origin.toLowerCase() === data.destination.toLowerCase()) {
  alert("Origin and destination must be different.");
  return;
```

```
  }

  if (id) {
    const t = trains.find(x => x.id === id);
    if (!t) return;

    const activeCount = reservations.filter(r =>
      r.trainId === id && r.status !== "Cancelled"
    ).length;

    if (data.seats < activeCount) {
      alert("Seat count cannot be less than the number of active reservations.");
      return;
    }

    Object.assign(t, data);
    alert("Train information updated.");
  } else {
    trains.push({id: nextTrainId++, ...data});
    alert("New train schedule added.");
  }

  clearTrainForm();
  renderAdmin();
  renderTrains(trains);
}

function editTrain(id) {
```

```
  if (!isAdmin()) return;

  const t = trains.find(x => x.id === id);
  if (!t) return;

  document.getElementById("editTrainId").value = t.id;
  document.getElementById("trainNumber").value = t.number;
  document.getElementById("trainName").value = t.name;
  document.getElementById("trainOrigin").value = t.origin;
  document.getElementById("trainDestination").value = t.destination;
  document.getElementById("trainTime").value = t.time;
  document.getElementById("trainDate").value = t.date;
  document.getElementById("trainSeats").value = t.seats;
  document.getElementById("fareSL").value = t.fares.SL;
  document.getElementById("fare3A").value = t.fares["3A"];
  document.getElementById("fare2A").value = t.fares["2A"];
  document.getElementById("fareCC").value = t.fares.CC;

  window.scrollTo(0, 0);
}

function clearTrainForm() {
  document.getElementById("editTrainId").value = "";
  document.querySelector("#adminTools form").reset();
  document.getElementById("trainDate").value = today;
}

function deleteTrain(id) {
```

```
  if (!isAdmin()) return;

  const active = reservations.some(r =>
    r.trainId === id && r.status !== "Cancelled"
  );

  if (active) {
    alert("This train has active reservations. Cancel those tickets before deleting it.");
    return;
  }

  if (confirm("Delete this train schedule?")) {
    trains = trains.filter(t => t.id !== id);
    renderAdmin();
    renderTrains(trains);
  }
}

function removeUser(index) {
  if (!isAdmin()) return;

  const user = users[index];
  if (!user || user.role === "admin") return;

  const hasActive = reservations.some(r =>
    r.userEmail === user.email && r.status !== "Cancelled"
  );
```

```
  if (hasActive) {
    alert("This user has active reservations. Cancel them before removing the user.");
    return;
  }

  if (confirm("Remove this registered user?")) {
    users.splice(index, 1);
    renderAdmin();
  }
}

function generateReport() {
  if (!isAdmin()) return;

  const active = reservations.filter(r => r.status !== "Cancelled");
  const cancelled = reservations.filter(r => r.status === "Cancelled");
  const fareTotal = active.reduce((sum, r) => sum + r.fare, 0);
  const refunds = cancelled.reduce((sum, r) => sum + r.refund, 0);

  document.getElementById("reportOutput").innerHTML = `
    <h3>Reservation Report</h3>
    <p>Total train schedules: ${trains.length}</p>
    <p>Registered passengers: ${users.filter(u => u.role === "user").length}</p>
    <p>Total tickets issued: ${reservations.length}</p>
    <p>Active reservations: ${active.length}</p>
    <p>Cancelled tickets: ${cancelled.length}</p>
    <p>Total active ticket fares: ₹${fareTotal}</p>
    <p>Calculated refunds: ₹${refunds}</p>
```

```
    <p class="small">Payment values are simulated and do not represent actual transactions.</p>`;
}

function downloadReport() {
  if (!isAdmin()) {
    alert("Admin login required.");
    return;
  }

  const rows = [
    ["PNR", "Passenger", "Email", "Train", "Origin", "Destination",
     "Date", "Class", "Seat", "Fare", "Payment Mode", "Status", "Refund"],
    ...reservations.map(r => [
      r.pnr, r.passengerName, r.email, r.trainName, r.origin,
      r.destination, r.date, r.travelClass, r.seat, r.fare,
      r.paymentMode, r.status, r.refund
    ])
  ];

  const csv = rows.map(row => row.map(v =>
    '"' + String(v).replace(/"/g, '""') + '"'
  ).join(",")).join("\n");

  const blob = new Blob([csv], {type: "text/csv;charset=utf-8;"});
  const url = URL.createObjectURL(blob);
  const link = document.createElement("a");
```

```
    link.href = url;
    link.download = "railway-reservation-report.csv";
    link.click();

    URL.revokeObjectURL(url);
}

function saveRefundRule() {
    if (!isAdmin()) {
        alert("Admin login required.");
        return;
    }

    const value = Number(document.getElementById("refundPercent").value);

    if (!Number.isFinite(value) || value < 0 || value > 100) {
        alert("Enter a refund percentage between 0 and 100.");
        return;
    }

    refundPercentage = value;
    alert("Demo refund rule updated to " + refundPercentage + "%.");
}

function contactUs(event) {
    event.preventDefault();
    alert("Thank you. This demo does not send messages to a server.");
    event.target.reset();
```

```
}

trains.forEach(t => {
  if (!t.date) t.date = today;
});

renderTrains(trains);
</script>

<style>
@media print {
  header, footer, nav, main > .page:not(#ticket),
  #ticket > button, #message {
    display: none !important;
  }

  body {
    background: white;
  }

  #ticket {
    display: block !important;
  }

  #ticket .card {
    box-shadow: none;
    border: 1px solid #777;
  }
```

```
}
</style>


</body>
</html>
```